Notes: Doshi, Jacobs, and Liu (JFE, 2024)

Modeling Volatility in Dynamic Term Structure Models

Contents

1 Big picture

- **Question.** Can a no-arbitrage dynamic term structure model match both the level of Treasury yields and the time variation in yield volatility?
- **Problem in standard models.** Canonical affine term structure models with stochastic volatility typically make volatility a linear function of the levels of the same factors that price the yield curve. This creates a tension: good yield fit often comes with weak volatility fit.
- **Main idea.** The paper proposes no-arbitrage term structure models in which volatility factors follow GARCH dynamics. Conditional variance is driven by lagged squared factor innovations rather than by factor levels.
- **Baseline model.** The empirical baseline uses three latent yield factors and one time-varying volatility factor. Only the volatility of the first state variable follows a GARCH process; the other two factor volatilities are constant.
- **Data.** The model is estimated using monthly zero-coupon Treasury yields from November 1971 to October 2019. The option-pricing application uses Treasury futures options from May 1988 to June 2016.
- **Benchmark models.** The paper compares the no-arbitrage GARCH model with canonical and restricted $A_1(3)$ affine stochastic volatility models.
- **Main volatility result.** The GARCH model has much higher correlations with model-free volatility benchmarks and much lower volatility RMSEs, especially at intermediate and long maturities.
- **Main yield result.** The improved volatility fit does not come at the cost of poor yield fit. Yield RMSEs are close to those from standard stochastic volatility term structure models.
- **Risk-premium result.** The no-arbitrage GARCH model matches the Campbell–Shiller expectations-hypothesis regression coefficients well, so it preserves key bond-risk-premium patterns.
- **Option-pricing result.** When parameters estimated from Treasury yields are used out of sample to price Treasury futures options, the GARCH model substantially outperforms the canonical $A_1(3)$ model.
- **Economic takeaway.** The specification of the volatility process is central. Modeling volatility as a function of lagged squared innovations can solve much of the volatility-fit problem without abandoning no-arbitrage tractability.

2 Data and measurement

2.1 Treasury yield data

- The sample contains monthly zero-coupon Treasury yields.
- Maturities are three months, six months, one year, two years, three years, four years, five years, and ten years.
- Three- and six-month yields come from FRED.
- One- through five-year and ten-year yields come from the Gurkaynak–Sack–Wright dataset.
- The sample period is November 1971 to October 2019.

Interpretation: The yield data are long enough to include the high-volatility monetary experiment of the early 1980s and the later low-volatility period. This makes the volatility-fit exercise demanding.

2.2 Model-free volatility benchmarks

The paper compares model-implied conditional yield volatility with two external benchmarks.

- **Realized volatility:** monthly realized variance constructed from within-month squared changes in daily yields.
- **EGARCH volatility:** EGARCH(1,1) volatility estimated assuming that monthly yield changes have an AR(1) conditional mean.

Interpretation: Realized volatility is noisy because it is an ex-post monthly object. EGARCH volatility is smoother because it is a statistical conditional expectation. A good term-structure model should be close to both.

2.3 Treasury futures options data

- Treasury futures options and underlying futures are from the CME.
- The options sample runs from May 1988 to June 2016.
- The paper studies options on five-year and ten-year Treasury futures.
- Contracts are grouped by maturity: less than 90 days, 90–182 days, and 182–270 days.
- Contracts are grouped by moneyness: 90–95, 95–97.5, 97.5–100, 100–102.5, 102.5–105, and 105–110.
- Option-implied volatilities are computed using the Black model.

Interpretation: The option-pricing exercise is an out-of-sample test: the model parameters are estimated from Treasury yields, not from option prices.

3 Models and empirical specifications

3.1 Baseline no-arbitrage GARCH term structure model

The model uses three latent state variables X_t. Under the physical and risk-neutral measures,

$$X_{t+1} = K_0^P + K_1^P X_t + \sqrt{\Sigma_{t+1}}\,\epsilon_{t+1}, \tag{1}$$
$$X_{t+1} = K_0^Q + K_1^Q X_t + \sqrt{\Sigma_{t+1}}\,\epsilon_{t+1}, \tag{2}$$

where $\epsilon_{t+1} \sim N(0, I)$ and Σ_{t+1} is diagonal.

The short rate is affine in the state vector:

$$r_t = \rho_0 + \rho_1 X_t. \tag{3}$$

Interpretation: The model remains a no-arbitrage dynamic term structure model, but its conditional covariance matrix is not linear in the levels of the factors.

3.2 GARCH volatility dynamics

The ith diagonal element of Σ_{t+1} follows a GARCH(1,1) process:

$$\sigma_{i,t+1}^2 = \omega_i + \beta_i \sigma_{i,t}^2 + \alpha_i \epsilon_{i,t}^2. \tag{4}$$

Equivalently, using filtered state innovations,

$$\sigma_{i,t+1}^2 = \omega_i + \beta_i \sigma_{i,t}^2 + \alpha_i \frac{\left(X_{i,t} - K_{0(i)}^P - K_{1(i,i)}^P X_{i,t-1}\right)^2}{\sigma_{i,t}^2}. \tag{5}$$

Interpretation: This is the paper's central modeling innovation. Volatility responds to squared shocks to yield factors, which is the GARCH logic, rather than responding mechanically to the level of the yield factors.

3.3 Pricing kernel and risk-neutral restrictions

The pricing kernel is

$$m_{t+1} = \exp\left(-r_t - \frac{1}{2}\lambda_t'\lambda_t - \lambda_t'\epsilon_{t+1}\right), \tag{6}$$

with an essentially affine price of risk,

$$\lambda_t = \left(\sqrt{\Sigma_{t+1}}\right)^{-1}(\lambda_0 + \lambda_1 X_t). \tag{7}$$

This implies

$$K_0^Q = K_0^P - \lambda_0, \qquad K_1^Q = K_1^P - \lambda_1. \tag{8}$$

Interpretation: The market price of risk links physical dynamics to risk-neutral pricing while preserving analytical bond prices.

3.4 Bond prices and yields

The zero-coupon bond price with maturity n has the exponential form

$$\widehat{P}_t^n = \exp\left(A_n(\Theta^Q) + B_n(\Theta^Q)'X_t + \sum_i C_{i,n}(\Theta^Q)\sigma_{i,t+1}^2\right). \tag{9}$$

The continuously compounded yield is therefore

$$\widehat{y}_t^n = A_n + B_n'X_t + \sum_i C_{i,n}\sigma_{i,t+1}^2. \tag{10}$$

Interpretation: The model is close in tractability to affine term structure models. The extra variance terms enter the yield formula analytically, which avoids simulation for bond pricing.

3.5 Benchmark affine stochastic volatility model

The benchmark $A_j(3)$ affine stochastic volatility model uses continuous-time factor dynamics with conditional variance that is affine in the state vector:

$$dX_t = K_{1\Delta}^Q(K_{0\Delta}^Q - X_t)dt + \sqrt{\Sigma_t}\, dW_t^Q, \tag{11}$$

with

$$\sigma_{i,t}^2 = a_i + b_i'X_t. \tag{12}$$

Interpretation: The key distinction is parametric. In standard affine volatility models, volatility is a linear function of factor levels. In the proposed model, volatility is a function of lagged squared innovations.

3.6 Model-implied conditional yield variance

The model-implied one-month conditional variance of an n-maturity yield is

$$\widehat{\mathrm{var}}_t(y_{t+1}^n) = B_n' \,\mathrm{var}_t(X_{t+1})B_n + \sigma_e^2, \tag{13}$$

where σ_e^2 is the variance of pricing errors.

Interpretation: This is the object compared with realized volatility and EGARCH volatility.

3.7 Campbell–Shiller expectations-hypothesis regression

The paper evaluates bond-risk-premium implications using

$$y_{t+1}^{n-1} - y_t^n = \psi_n + \varphi_n \left(\frac{y_t^n - y_t^1}{n-1} \right) + e_{t+1}^n. \tag{14}$$

Interpretation: Under the expectations hypothesis, $\varphi_n = 1$. In the data, coefficients are negative and more negative at longer maturities. A useful term-structure model should reproduce this pattern.

3.8 Option-implied volatility interpolation

For the option application, data-based at-the-money implied volatility is constructed using

$$\sigma(m, T) = a + bm + cT + dm^2 + eT^2 + fmT + \varepsilon, \tag{15}$$

where m is moneyness and T is maturity.

Interpretation: The interpolation creates a comparable 90-day at-the-money implied volatility series across months and across five-year and ten-year Treasury futures options.

4 Identification and empirical comparison design

4.1 What identifies the model improvement

- All models are estimated on the same monthly yield data.
- Models are compared using the same volatility benchmarks and the same yield-fitting criteria.
- The restricted $A_1(3)$ benchmark makes state-variable dynamics as close as possible to the GARCH model, so the main remaining difference is the volatility specification.
- Treasury futures options are used as an out-of-sample test because option payoffs are very sensitive to volatility.

Interpretation: The empirical comparison is not based on a single in-sample objective. The GARCH specification must improve volatility fit without harming yield fit and must help price derivatives.

4.2 Why the early 1980s matter

- The early 1980s were a period of exceptionally high interest-rate volatility.
- Standard affine volatility models tend to have difficulty fitting both the early-1980s spike and the later low-volatility period.
- The GARCH model captures the spike better at intermediate and long maturities because volatility reacts to large factor innovations.

Interpretation: This is the central time-series stress test. A volatility model that cannot handle both regimes will either overstate calm-period volatility or understate crisis-period volatility.

4.3 What the paper cannot fully prove

- The paper compares parametric models; it does not prove that GARCH is the unique best way to model yield volatility.
- The option-pricing errors remain large, so the model is not a complete option-pricing model.

- The options exercise uses yield-estimated parameters rather than jointly estimating yields and options.
- The baseline model is deliberately parsimonious, with only one time-varying volatility factor.

Interpretation: The contribution is not that all term-structure volatility problems are solved. The contribution is that a simple GARCH volatility factor produces a major improvement inside a tractable no-arbitrage framework.

5 Tables and figures

The figures and tables below are directly cropped from the original paper and grouped to save space.

5.1 Table 1 + Table 2: Yield and option data

Read:

- Table 1 shows that yield levels are persistent and upward sloping on average.
- Realized and EGARCH volatilities are highly skewed and leptokurtic, consistent with volatility spikes.
- Table 2 shows option-implied volatilities by moneyness and maturity for five-year and ten-year Treasury futures options.
- Ten-year futures options have higher average implied volatilities than five-year futures options.

Economic message: The data contain both yield-level information and volatility information. The model must fit the yield curve while also matching volatile conditional second moments.

Table 1
Summary statistics for yields and volatilities.

Panel A: Yields

	Central moments				Autocorrelation		
	Mean (%)	St.Dev (%)	Skewness	Kurtosis	Lag 1	Lag 12	Lag 30
3 month	4.6244	3.4488	0.6335	3.3177	0.9906	0.8640	0.6455
6 month	4.7501	3.4363	0.5503	3.0660	0.9916	0.8755	0.6693
1 year	5.1127	3.5684	0.4668	2.8273	0.9906	0.8879	0.7074
2 year	5.3417	3.5021	0.3936	2.6794	0.9915	0.9020	0.7542
3 year	5.5349	3.4155	0.3768	2.6240	0.9921	0.9090	0.7809
4 year	5.7027	3.3274	0.3806	2.6082	0.9924	0.9126	0.7968
5 year	5.8501	3.2450	0.3916	2.6131	0.9924	0.9140	0.8060
10 year	6.3594	2.9598	0.4340	2.7129	0.9920	0.9115	0.8141

Panel B: Realized Volatilities

	Central moments				Autocorrelation		
	Mean (bps)	St.Dev (bps)	Skewness	Kurtosis	Lag 1	Lag 12	Lag 30
1 year	23.2313	19.1239	2.5249	11.2102	0.7363	0.4920	0.1442
2 year	26.1794	17.6976	2.4249	11.0406	0.7138	0.4571	0.0607
3 year	27.4345	16.7397	2.3646	10.8820	0.7140	0.4260	0.0238
4 year	27.6734	15.8503	2.2507	10.4456	0.7240	0.4120	0.0180
5 year	27.5106	15.1577	2.1224	9.8979	0.7306	0.4141	0.0288
10 year	26.4272	13.8500	1.7106	7.4772	0.7537	0.4690	0.0819

Panel C: EGARCH Volatilities

	Central moments				Autocorrelation		
	Mean (bps)	St.Dev (bps)	Skewness	Kurtosis	Lag 1	Lag 12	Lag 30
3 month	29.5289	33.0125	2.8007	11.9469	0.9699	0.6995	0.2779
6 month	28.6691	29.0062	2.6993	12.2778	0.9665	0.6705	0.3072
1 year	35.4025	28.5081	2.3117	10.3576	0.9611	0.7131	0.4225
2 year	35.9635	22.0695	2.1404	9.2477	0.9722	0.7253	0.4274
3 year	35.0766	16.5472	1.9735	8.2257	0.9721	0.7203	0.4065
4 year	34.0902	13.1045	1.8018	7.1088	0.9710	0.7160	0.3861
5 year	33.2059	11.1209	1.6611	6.1600	0.9719	0.7264	0.3803
10 year	30.3622	9.0130	1.3719	4.6081	0.9655	0.6987	0.3660

Notes to Table: We present summary statistics for the data used in estimation. We present the sample mean, standard deviation, skewness, kurtosis, and autocorrelations for each of the yields (Panel A) and yield volatilities (Panels B and C). The yields are continuously compounded monthly zero-coupon bond yields. The monthly realized volatility in Panel B is the square root of monthly realized variance. We construct monthly realized variance using within-month squared changes in daily yields. Panel C reports on the EGARCH(1, 1) estimates of changes in monthly yields. The EGARCH(1, 1) is estimated assuming that the conditional mean of changes in monthly yields is generated by an AR(1) process. The sample period is from 1971:11 to 2019:10.

ds are from the Gürkaynak et al. (2007, GSW) dataset. The *3.2. Treasury options data*

(a) Summary statistics for yields and volatilities

Table 2
Summary statistics for futures options.

Panel A: Implied Volatility: Options on Five-Year Treasury Futures

	Moneyness (K/F)							
Maturity	90–95	95–97.5	97.5–100	100–102.5	102.5–105	105–110	All	N
<90	6.84%	4.41%	3.00%	2.83%	4.01%	6.51%	3.81%	5811
90–182	5.41%	4.29%	3.53%	3.37%	3.97%	5.18%	3.79%	6429
182–270	5.57%	4.47%	3.83%	3.67%	3.71%	4.07%	3.86%	461
All	6.26%	4.35%	3.32%	3.15%	3.98%	5.91%	3.80%	
N	949	2063	4333	3731	1229	396		

Panel B: Implied Volatility: Options on Ten-Year Treasury Futures

	Moneyness (K/F)							
Maturity	90–95	95–97.5	97.5–100	100–102.5	102.5–105	105–110	All	N
<90	8.12%	5.50%	4.55%	4.31%	5.06%	7.19%	5.79%	6707
90–182	7.13%	6.00%	5.56%	5.39%	5.52%	6.39%	5.97%	9531
182–270	6.79%	6.03%	5.79%	5.68%	5.68%	5.80%	5.94%	2583
All	7.48%	5.83%	5.27%	5.07%	5.37%	6.64%	5.90%	
N	3395	3012	3903	3661	2531	2319		

Notes to Table: We present summary statistics for option-implied volatilities for a sample of Treasury futures options. The sample period is from May 1988 to June 2016. Panels A and B report the average implied volatilities from futures options with underlying five- and ten-year Treasuries, for various moneyness and maturity groups. N indicates the number of option contracts in each group.

(b) Summary statistics for Treasury futures options

5.2 Table 3: Parameter estimates

Read:

- The no-arbitrage GARCH model, canonical $A_1(3)$ model, and restricted $A_1(3)$ model are estimated on the same monthly zero-coupon yields.
- In the GARCH model, the volatility persistence parameter is high, with β around 0.918.
- The canonical and restricted $A_1(3)$ models use a level factor to drive volatility.

Economic message: The estimates implement the key comparison: similar term-structure frameworks, but different volatility dynamics.

Table 3
Parameter estimates.

Panel A: The No-Arbitrage GARCH Model								
K_0^P	K_1^P				K_0^Q	K_1^Q		
0.0036	0.9996				0.0009	0.9963		
−0.0029		0.9855			−0.0064		0.9611	
0.0019			0.8826		0.0031			0.8295
ρ_0		ρ_1				Log Likelihood		
−0.0002		0.0004	−0.0491	0.1074		21987.48		
$\omega \times 1e4$	α	β		$\sigma_2^2 \times 1e4$	$\sigma_3^2 \times 1e4$			
0.0008	0.0530	0.9182		0.1623	0.2055			
Panel B: The Canonical $A_1(3)$ Model								
K_0^P	K_1^P				K_0^Q	K_1^Q		
0.0708	0.9763				0.0708	1.0003		
0.0002	−0.0001	0.9732	−0.0001		0.0002	−0.0012	0.9684	−0.0002
0.0002	−0.0001	−0.0001	0.7978		0.0002	−0.0002	−0.0002	0.8379
ρ_0		ρ_1				Log Likelihood		
−0.0012		0.0036	0.0016	0.0134		22239.58		
a				b				
0.0000	1.0000	1.0000		1.0000	12.8712	0.2459		
Panel C: The Restricted $A_1(3)$ Model								
K_0^P	K_1^P				K_0^Q	K_1^Q		
0.1026	0.9808				0.1026	1.0002		
0.0000		0.9716			0.0000		0.9711	
0.0000			0.7879		0.0000			0.8525
ρ_0		ρ_1				Log Likelihood		
−0.0022		0.0022	−0.0017	0.0133		22206.68		
a				b				
0.0000	1.0000	1.0000		1.0000	10.0435	0.1874		

Notes to Table: We present the estimated parameters and log likelihoods for the no-arbitrage GARCH model (Panel A), the canonical $A_1(3)$ model (Panel B), and the restricted $A_1(3)$ model (Panel C). In the restricted $A_1(3)$ model, we set the feedback matrix to be a diagonal matrix. The estimates are based on monthly zero-coupon bond yields from 1971:11 to 2019:10. The state variables in the canonical and restricted $A_1(3)$

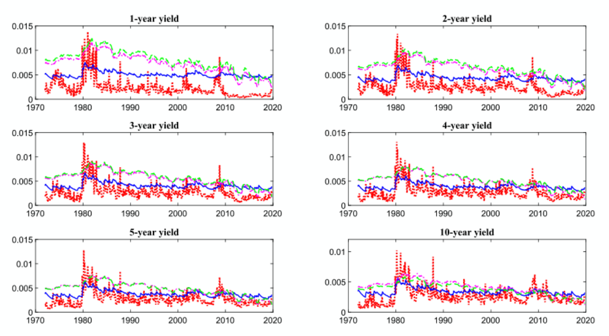
Figure 2: Parameter estimates for the no-arbitrage GARCH and benchmark models

5.3 Figures 1–2: Conditional volatility versus model-free benchmarks

Read:

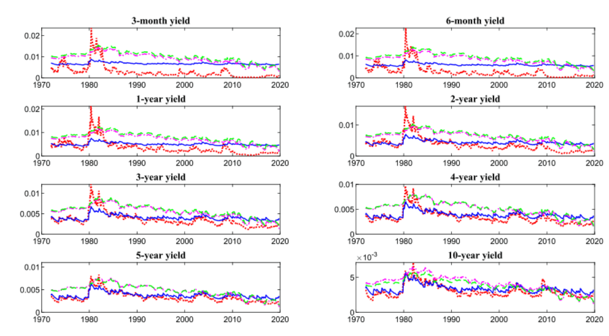
- Figure 1 compares model-implied conditional volatility with realized volatility.
- Figure 2 compares model-implied conditional volatility with EGARCH volatility.
- The no-arbitrage GARCH model tracks the early-1980s volatility spike more closely at longer maturities.
- The benchmark $A_1(3)$ models tend to overstate volatility in calmer periods and miss some of the time-series behavior.

Economic message: This is the paper's main visual evidence. Lagged squared innovations produce a volatility process that better follows the true volatility path.



odel-implied conditional volatility by maturity: Comparison with realized volatility.
igure: We plot the implied one-month conditional yield volatility for different models. For each maturity, the dotted line (red) represents "model-free" volatility, n
are root of realized variance. The maturity-specific monthly realized variances are computed using within-month squared changes in daily yields. The solid lir
the conditional volatility from the no-arbitrage GARCH model. The dash–dot line (magenta) represents the conditional volatility from the canonical $A_1(3)$ mo
e (green) represents the conditional volatility from the restricted $A_1(3)$ model. In the restricted $A_1(3)$ model, we set the feedback matrix to be a diagonal mat
tion of the references to color in this figure legend, the reader is referred to the web version of this article.)

(a) Comparison with realized volatility



odel-implied conditional volatility by maturity: Comparison with EGARCH volatility.
igure: We plot the implied one-month conditional yield volatility for different models. For each maturity, the dotted line (red) represents "model-free" volatility, n
ARCH(1, 1) volatility. The maturity-specific EGARCH(1, 1) is estimated assuming that the conditional mean of changes in monthly yields is generated by an AR(1)
line (blue) represents the conditional volatility from the no-arbitrage GARCH model. The dash–dot line (magenta) represents the conditional volatility from the c
el. The dashed line (green) represents the conditional volatility from the restricted $A_1(3)$ model. In the restricted $A_1(3)$ model, we set the feedback matrix to be a
or interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(b) Comparison with EGARCH volatility

5.4 Table 4 + Table 5: Correlations and RMSEs for volatility fit

Read:

- Table 4 shows that average correlations with realized volatility are 0.65 for GARCH and 0.47 for the benchmark models.
- Average correlations with EGARCH volatility are 0.80 for GARCH and 0.68 for the benchmark models.
- Table 5 shows that GARCH reduces average realized-volatility RMSE to 19.74 bps, compared with 33.01 and 34.91 bps for the two benchmarks.
- Against EGARCH volatility, GARCH has an average RMSE of 21.39 bps, compared with 34.21 and 37.50 bps.

Economic message: The GARCH model improves volatility fit by roughly 40 percent relative to benchmark stochastic volatility term-structure models.

Table 4

Unconditional correlations for conditional volatility implied by different models.

Panel A: Unconditional Correlations with Realized Volatility

	1 year	2 year	3 year	4 year	5 year	10 year	Avg.
No-Arbitrage GARCH	0.56	0.61	0.65	0.68	0.69	0.71	0.65
Canonical $A_1(3)$	0.54	0.51	0.48	0.46	0.43	0.38	0.47
Restricted $A_1(3)$	0.54	0.51	0.49	0.46	0.44	0.39	0.47

Panel B: Unconditional Correlations with EGARCH(1, 1) Volatility

	3 month	6 month	1 year	2 year	3 year	4 year	5 year	10 year	Avg.
No-Arbitrage GARCH	0.65	0.66	0.70	0.79	0.85	0.90	0.92	0.91	0.80
Canonical $A_1(3)$	0.58	0.60	0.68	0.73	0.74	0.74	0.72	0.61	0.68
Restricted $A_1(3)$	0.57	0.60	0.68	0.73	0.75	0.75	0.73	0.62	0.68

Notes to Table: Panel A presents the unconditional correlations between the realized volatility and the one-month conditional volatility implied by the no-arbitrage GARCH model, the canonical $A_1(3)$ model, and the restricted $A_1(3)$ model. We construct monthly realized variance using within-month squared changes in daily yields. Panel B presents the unconditional correlations between the EGARCH(1, 1) volatility and the one-month conditional volatility implied by the three models. The EGARCH(1, 1) is estimated assuming that the conditional mean of changes in monthly yields is generated by an AR(1) process. In the restricted $A_1(3)$ model, we set the feedback matrix to be a diagonal matrix. The last column shows the average correlation across all maturities.

(a) Unconditional correlations for conditional volatility

Table 5

Conditional volatility fit.

Panel A: RMSEs of Volatility Based on Realized Volatility

	1 year	2 year	3 year	4 year	5 year	10 year	Avg.
No-Arbitrage GARCH	31.85	22.79	18.85	16.69	15.40	12.86	19.74
Canonical $A_1(3)$	53.12	38.94	32.03	28.07	25.58	20.33	33.01
Restricted $A_1(3)$	59.82	43.16	34.37	28.97	25.38	17.76	34.91
Improvement on Can. $A_1(3)$	40.04%	41.47%	41.16%	40.52%	39.80%	36.74%	39.95%
Improvement on Rest. $A_1(3)$	46.76%	47.20%	45.16%	42.37%	39.32%	27.58%	41.40%

Panel B: RMSEs of Volatility Based on EGARCH(1, 1) Volatility

	3 month	6 month	1 year	2 year	3 year	4 year	5 year	10 year	Avg.
No-Arbitrage GARCH	48.36	40.05	28.92	19.04	12.77	9.15	7.23	5.57	21.39
Canonical $A_1(3)$	68.31	59.02	42.92	29.32	22.94	19.53	17.54	14.09	34.21
Restricted $A_1(3)$	76.39	66.69	49.18	33.38	25.33	20.49	17.31	11.19	37.50
Improvement on Can. $A_1(3)$	29.21%	32.15%	32.60%	35.06%	44.34%	53.14%	58.80%	60.45%	43.22%
Improvement on Rest. $A_1(3)$	36.69%	39.95%	41.19%	42.96%	49.59%	55.33%	58.25%	50.23%	46.77%

Notes to Table: We present the volatility fit for the no-arbitrage GARCH model, the canonical $A_1(3)$ model, and the restricted $A_1(3)$ model. In the restricted $A_1(3)$ model, we set the feedback matrix to be a diagonal matrix. In Panel A, the RMSEs are computed by using realized volatility as a measure of model-free volatility. We construct monthly realized variance using within-month squared changes in daily yields. In Panel B, the RMSEs are computed by using an EGARCH(1, 1) model as a measure of model-free volatility. The EGARCH(1, 1) is estimated assuming that the conditional mean of changes in monthly yields is generated by an AR(1) process. RMSEs are reported in basis points. We also present the

(b) RMSEs of conditional volatility fit

5.5 Figure 3 + Table 6: Unconditional volatility and yield fit

Read:

- Figure 3 shows the term structure of unconditional yield volatility.
- The GARCH-implied term structure is closer to the realized and EGARCH benchmarks than the $A_1(3)$ benchmarks.
- Table 6 shows that the GARCH model has average yield RMSE of 9.88 bps, only modestly above the benchmark stochastic-volatility models.
- Gaussian models fit yield levels somewhat better, but they lack time-varying volatility.

Economic message: The improved volatility fit is not purchased by sacrificing the yield curve. The no-arbitrage GARCH model gives a favorable trade-off between first and second moments.

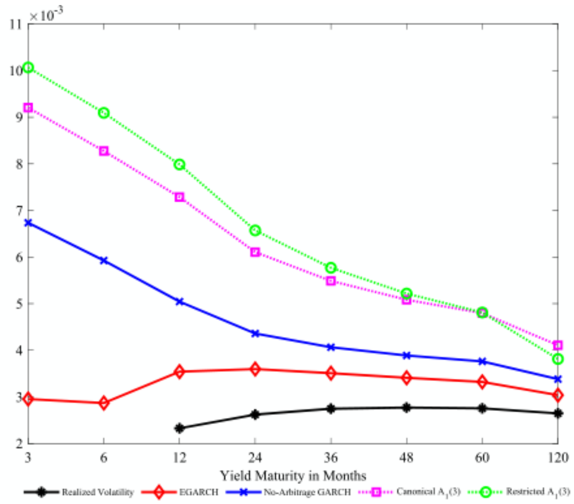


Fig. 3. The term structure of unconditional volatility.
Notes to Figure: We plot the term structure of average yield volatility implied by different models together with the term structure of "model-free" realized volatility and EGARCH volatility. The solid line with asterisk (black) represents the term structure of realized volatility, and the diamond line (red) represents the EGARCH(1, 1) term structure. The cross line (blue) represents the term structure of volatility from the no-arbitrage GARCH model. The square line (magenta) represents the term structure of volatility from the canonical $A_1(3)$ model. The circle line (green) represents the term structure of volatility from the restricted $A_1(3)$ model. In the restricted $A_1(3)$ model, we set the feedback matrix to be a diagonal matrix. The x-axis is yield maturity in months. (For interpretation of the references to color in this figure legend, the reader

(a) Term structure of unconditional volatility

Table 6
Yield fit.

Panel A: Stochastic Volatility Models	3 month	6 month	1 year	2 year	3 year	4 year	5 year	10 year	Avg.
No-Arbitrage GARCH	8.13	11.89	16.21	8.10	6.38	6.61	6.95	14.81	9.88
Canonical $A_1(3)$	7.47	13.17	16.02	4.95	4.94	5.94	6.06	9.22	8.47
Restricted $A_1(3)$	7.82	12.85	16.36	4.88	4.72	5.82	6.22	8.48	8.39
Panel B: Gaussian Models	3 month	6 month	1 year	2 year	3 year	4 year	5 year	10 year	Avg.
JSZ	7.93	10.30	12.84	4.63	5.06	5.63	5.93	7.68	7.50
JSZ Restricted_1	7.38	10.83	13.61	4.35	4.95	5.88	5.74	7.76	7.56
JSZ Restricted_2	5.72	13.09	15.28	5.81	4.81	6.94	7.86	7.82	8.42
JSZ Restricted_3	9.41	12.48	14.13	6.90	6.28	7.12	7.48	9.00	9.10

Notes to Table: We present RMSEs based on the yield fit for the time-varying volatility models (Panel A) and the Gaussian models (Panel B). Panel A reports results for the no-arbitrage GARCH model, the canonical $A_1(3)$ model, and the restricted $A_1(3)$ model. In the restricted $A_1(3)$ model, we set the feedback matrix to be a diagonal matrix. In Panel B, we consider the canonical representation of Joslin, Singleton, and Zhu (2011, referred to as JSZ). We present the results for the maximum flexible specification of the JSZ model and also for three restricted JSZ canonical forms. In the first restricted version, we set the variance–covariance matrix to be a diagonal matrix. In the second restricted version, we also set the $(1, 2)$ and $(1, 3)$ entries of the feedback matrix to be zeros. In the third restricted version, we set the feedback matrix to be a diagonal matrix. The last column in the table shows the average across all maturities. RMSEs are reported in basis points.

(b) Yield-fit RMSEs

5.6 Figure 4 + Table 7: Expectations hypothesis and expected realized variance

Read:

- Figure 4 shows that the GARCH model matches the negative Campbell–Shiller coefficients observed in the data.
- Table 7 uses an expected realized-variance benchmark instead of raw realized volatility.
- The GARCH model again outperforms the benchmark models, with average correlation 0.79 and average RMSE 16.52 bps.

Economic message: The GARCH model preserves bond-risk-premium patterns while improving volatility fit.

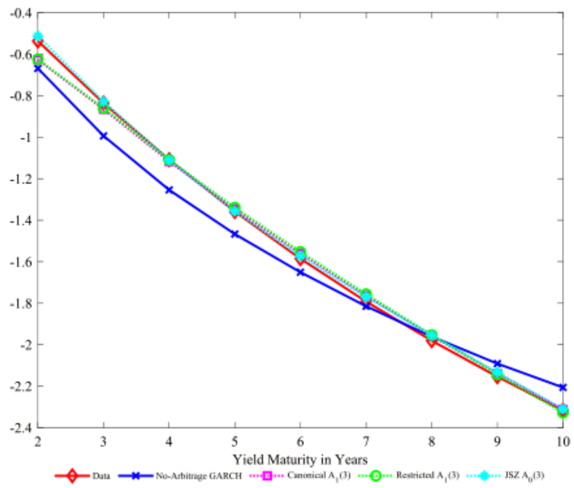


Fig. 4. The Campbell–Shiller expectations hypothesis regression.
Notes to Figure: We plot the estimated coefficients from the Campbell–Shiller expectation hypothesis regression using yields implied by different models. The diamond line (red) represents the estimates using the yield data. The cross line (blue) represents the estimates from yields implied by the no-arbitrage GARCH model. The square line (magenta) shows the estimates from yields implied by the canonical $A_1(3)$ model. The circle line (green) is for the estimates from yields implied by the restricted $A_1(3)$ model. In the restricted $A_1(3)$ model, we set the feedback matrix to be a diagonal matrix. The asterisk line (cyan) is for the estimates using yields from the Joslin, Singleton, and Zhu (2011) specification of the Gaussian $A_0(3)$ model. The x-axis is yield maturity in

(a) Campbell–Shiller expectations-hypothesis regression

Table 7
Conditional volatility fit based on expected realized variance.

Panel A: Unconditional Correlations							
	1 year	2 year	3 year	4 year	5 year	10 year	Avg.
No-Arbitrage GARCH	0.69	0.75	0.80	0.83	0.85	0.84	0.79
Canonical $A_1(3)$	0.64	0.61	0.57	0.53	0.50	0.43	0.55
Restricted $A_1(3)$	0.64	0.61	0.58	0.54	0.51	0.44	0.55
Panel B: RMSEs of Volatility							
	1 year	2 year	3 year	4 year	5 year	10 year	Avg.
No-Arbitrage GARCH	28.86	19.34	15.26	13.26	12.14	10.28	16.52
Canonical $A_1(3)$	50.17	35.78	28.89	25.16	22.92	18.26	30.20
Restricted $A_1(3)$	56.92	40.04	31.24	26.04	22.69	15.71	32.11
Improvement on Can. $A_1(3)$	42.48%	45.95%	47.16%	47.31%	47.03%	43.69%	45.60%
Improvement on Rest. $A_1(3)$	49.31%	51.70%	51.13%	49.09%	46.47%	34.58%	47.05%

Notes to Table: We present the unconditional correlations and the volatility fit for various models when the conditional volatility implied by the realized variance model is used as a measure of model-free yield volatility. We construct the measure of monthly variance using within-month squared changes in daily yields. The logarithm of the realized variance follows an ARMA(1, 1) process. Panel A presents the unconditional correlations between the one-month conditional volatility from the realized variance model and the one-month conditional volatility implied by the no-arbitrage GARCH model, the canonical $A_1(3)$ model, and the restricted $A_1(3)$ model. In the restricted $A_1(3)$ model, we set the feedback matrix to be a diagonal matrix. Panel B reports the volatility RMSEs in basis points for the three models. We also present the percentage improvement in RMSE for the no-arbitrage GARCH model over the canonical and restricted $A_1(3)$ models. The last column shows the averages

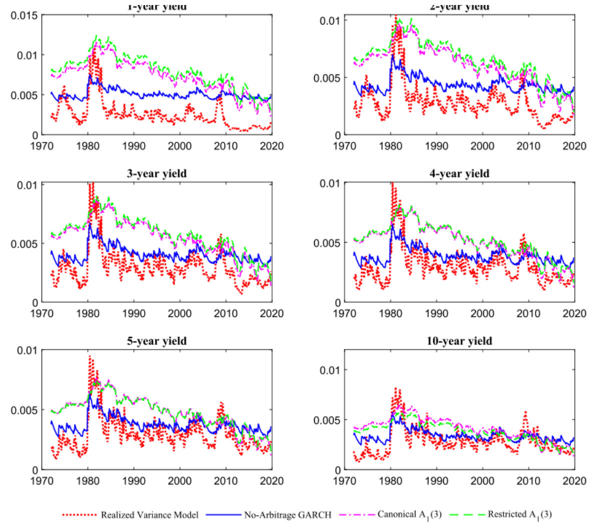
(b) Volatility fit based on expected realized variance

5.7 Figure 5 + Table 8: Alternative realized-variance benchmark and multiple volatility factors

Read:

- Figure 5 shows conditional volatility using the expected realized-variance benchmark.
- The GARCH model remains closer to the true volatility path than the benchmark $A_1(3)$ models.
- Table 8 estimates models with two and three volatility factors.
- Likelihood-ratio tests strongly favor adding volatility factors.

Economic message: More volatility factors improve likelihood and volatility fit, but the single-factor GARCH model already captures the main economic mechanism.



odel-implied conditional volatility: comparison with the expected realized variance.

(a) Conditional volatility versus expected realized variance

Table 8
Parameter estimates for the no-arbitrage GARCH model with multiple volatility factors.

Panel A: Two Volatility Factors								
K_0^P	K_1^P				K_0^Q	K_1^Q		
0.0031	0.9996				0.0006	0.9962		
−0.0026		0.9831			−0.0054		0.9639	
0.0019			0.9041		0.0033			0.8409
ρ_0		ρ_1				Log Likelihood		
−0.0002		0.0003	−0.0470	0.1156		22226.95		
$\omega \times 1e4$	$\alpha \times 1e4$	β		$\sigma_3^2 \times 1e4$				
0.0094	517.1944	0.9258		0.2124				
0.0091	0.0385	0.8000						
Panel B: Three Volatility Factors								
K_0^P	K_1^P				K_0^Q	K_1^Q		
0.0031	0.9996				0.0006	0.9962		
−0.0026		0.9831			−0.0054		0.9639	
0.0019			0.9041		0.0033			0.8409
ρ_0		ρ_1				Log Likelihood		
−0.0002		0.0003	−0.0470	0.1156		22439.59		
$\omega \times 1e4$	$\alpha \times 1e4$	β						
0.0094	517.1944	0.9258						
0.0454	0.0862	0.8000						
0.0179	0.0425	0.7000						
Panel C: Likelihood Ratio Test								
	Three vs One Volatility Factor				Two vs One Volatility Factor			
P Value	0.0000				0.0000			
Statistics	904.2304				478.9404			
Critical Value	9.4877				5.9915			

Notes to Table: We present the estimated parameters and log likelihoods for the no-arbitrage GARCH model with two volatility factors (Panel A) and the no-arbitrage GARCH model with three volatility factors (Panel B). The estimates are based on monthly zero coupon bond yields from 1971:11 to 2019:10. Panel C presents the p-value, the likelihood ratio test statistics, and the critical values for the no-arbitrage GARCH model, testing between the models with three and one volatility factors, and between the models with two and one volatility factors.

ould lead to more accurate option prices and superior *6.2. Modeling time-variation in treasury futures options prices*

(b) Parameter estimates for multiple-volatility-factor GARCH models

5.8 Table 9 + Figure 6: Multifactor GARCH performance

Read:

- Table 9 compares two-volatility-factor, three-volatility-factor, and principal-component GARCH models.
- Three principal components deliver high correlations with EGARCH volatility, averaging 0.89.
- Figure 6 shows that multifactor GARCH models better capture early-1980s volatility spikes.
- The improvement is strongest for longer-maturity yields.

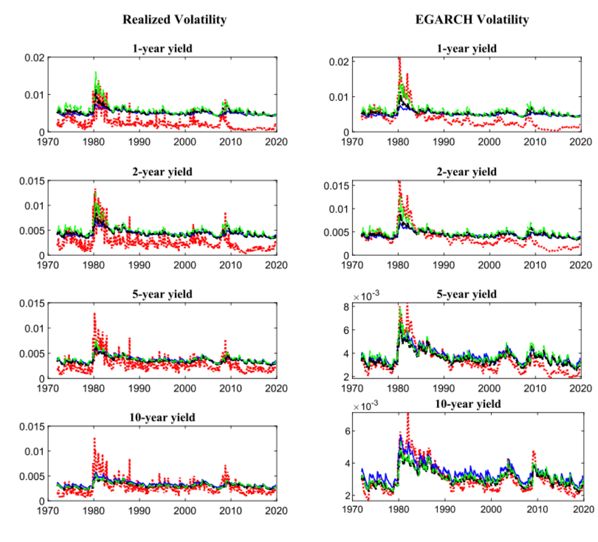
Economic message: Additional GARCH volatility factors refine the fit, but the key improvement is still the GARCH-style shock-driven variance process.

Table 9
Results for the no-arbitrage GARCH models with multiple volatility factors and with three principal components.

Panel A: Unconditional Correlations with Realized Volatility								
			1 year	2 year	3 year	4 year	5 year	10 year	Avg.
Two Volatility Factors			0.62	0.63	0.65	0.67	0.68	0.70	0.66
Three Volatility Factors			0.68	0.66	0.66	0.67	0.69	0.70	0.68
Three PCs			0.72	0.70	0.69	0.67	0.66	0.62	0.68
Panel B: RMSEs of Volatility Based on Realized Volatility									
			1 year	2 year	3 year	4 year	5 year	10 year	Avg.
Two Volatility Factors			34.04	23.04	18.31	15.78	14.28	11.65	19.52
Three Volatility Factors			38.87	26.84	20.93	17.59	15.54	11.91	21.95
Three PCs			26.47	23.06	21.30	20.40	19.93	18.91	21.68
Improvement on Can. $A_1(3)$									
Two Volatility Factors			35.92%	40.82%	42.85%	43.76%	44.19%	42.71%	41.71%
Three Volatility Factors			26.82%	31.07%	34.65%	37.33%	39.25%	41.43%	35.09%
Three PCs			50.17%	40.77%	33.49%	27.32%	22.10%	6.98%	30.14%
Improvement on Rest. $A_1(3)$									
Two Volatility Factors			43.10%	46.61%	46.74%	45.52%	43.75%	34.41%	43.36%
Three Volatility Factors			35.02%	37.82%	39.10%	39.29%	38.77%	32.95%	37.16%
Three PCs			55.75%	46.57%	38.02%	29.59%	21.48%	−6.48%	30.82%
Panel C: Unconditional Correlations with EGARCH(1, 1) Volatility									
	3 month	6 month	1 year	2 year	3 year	4 year	5 year	10 year	Avg.
Two Volatility Factors	0.79	0.79	0.79	0.81	0.83	0.85	0.88	0.90	0.83
Three Volatility Factors	0.86	0.85	0.84	0.84	0.84	0.85	0.86	0.89	0.85
Three PCs	0.84	0.85	0.89	0.93	0.94	0.94	0.92	0.84	0.89
Panel D: RMSEs of Volatility Based on EGARCH(1, 1) Volatility									
	3 month	6 month	1 year	2 year	3 year	4 year	5 year	10 year	Avg.
Two Volatility Factors	51.17	42.17	28.75	18.26	12.26	8.77	6.78	4.89	21.63
Three Volatility Factors	51.52	43.96	29.92	19.46	13.51	9.93	7.77	5.01	22.64
Three PCs	31.57	28.61	24.44	18.07	13.93	12.22	11.77	12.45	19.13
Improvement on Can. $A_1(3)$									
Two Volatility Factors	25.08%	28.56%	33.01%	37.72%	46.56%	55.12%	61.33%	65.30%	44.08%
Three Volatility Factors	24.58%	25.52%	30.29%	33.62%	41.10%	49.16%	55.68%	64.43%	40.55%
Three PCs	53.79%	51.52%	43.05%	38.37%	39.28%	37.43%	32.93%	11.59%	38.49%
Improvement on Rest. $A_1(3)$									
Two Volatility Factors	33.01%	36.78%	41.54%	45.30%	51.60%	57.21%	60.81%	56.33%	47.82%
Three Volatility Factors	32.56%	34.09%	39.17%	41.70%	46.65%	51.53%	55.08%	55.24%	44.50%
Three PCs	58.67%	57.10%	50.30%	45.87%	45.00%	40.35%	32.03%	−11.25%	39.76%

Notes to Table: We present the unconditional correlations and the volatility fit for the no-arbitrage GARCH models with two volatility factors, with three volatility factors, and with three principal components as factors, in which only the first principal component has a time-varying variance. Panel A presents the unconditional correlations between the realized volatility and the one-month conditional volatility implied by the three models. We construct monthly realized variance using within-month squared changes in daily yields. Panel C present the unconditional correlations between the EGARCH(1, 1) volatility and the one-month conditional volatility implied by the three models. The EGARCH(1, 1) is estimated assuming that the conditional mean of changes in monthly yields is generated by an AR(1) process. Panels B and D present the volatility fit for the three models. In Panel B, the RMSEs are computed by using realized volatility as a measure of model-free volatility. In Panel D, the RMSEs are computed by using an EGARCH(1, 1) model as a measure of model-free volatility. We also present the percentage RMSE improvement of the three models over the canonical and restricted $A_1(3)$ models. RMSEs are reported in basis points. The last column shows the averages across all maturities.

(a) Performance of multifactor GARCH models



›del-implied conditional volatility in the no-arbitrage GARCH model with multiple volatility factors.
gure: We plot the one-month conditional volatility of the yield curve implied by the no-arbitrage GARCH model with different numbers of volatility factors toge
l volatility (left four panels) and the EGARCH(1, 1) volatility (right four panels) respectively. The dotted line (red) in the left panels represents the realized

(b) Conditional volatility with multiple volatility factors

5.9 Table 10 + Figure 7: Other benchmarks and Treasury futures options

Read:

- Table 10 shows that the no-arbitrage $A_1(4)$ GARCH extension performs well across volatility and yield dimensions.
- The GARCH models continue to outperform quadratic and other benchmark models in volatility fit.
- Figure 7 compares model-implied at-the-money option volatility with data for five-year and ten-year Treasury futures options.
- The canonical $A_1(3)$ model overprices volatility much more strongly than the GARCH model.

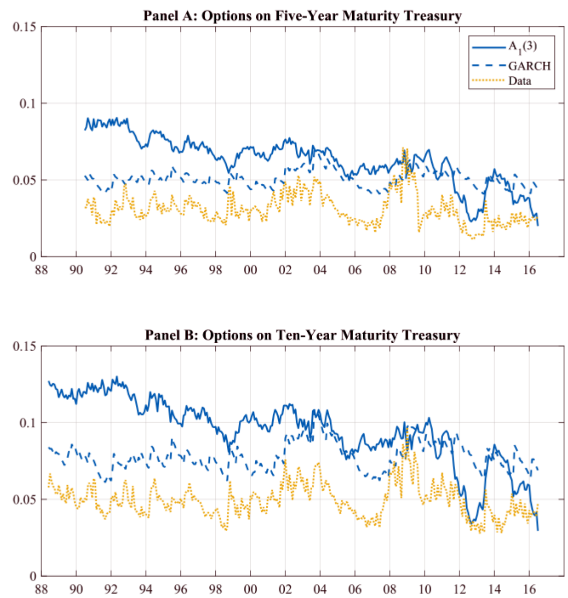
Economic message: The GARCH volatility specification improves derivative pricing because options are especially sensitive to conditional variance.

Table 10
Additional benchmark models.

	3 month	6 month	1 year	2 year	3 year	4 year	5 year	10 year	Avg.
Panel A: Unconditional Correlations with Realized Volatility									
No-Arbitrage $A_0(3)$ GARCH			0.56	0.61	0.65	0.68	0.69	0.71	0.65
JK $A_1(4)$ Model			0.51	0.45	0.39	0.33	0.29	0.24	0.37
Quadratic Model			0.57	0.50	0.44	0.40	0.36	0.32	0.43
No-Arbitrage $A_1(4)$ GARCH			0.76	0.73	0.69	0.66	0.63	0.54	0.67
Panel B: RMSEs of Volatility Based on Realized Volatility									
No-Arbitrage $A_0(3)$ GARCH			31.85	22.79	18.85	16.69	15.40	12.86	19.74
JK $A_1(4)$ Model			20.03	18.08	17.64	17.25	16.78	15.43	17.54
Quadratic Model			36.42	26.65	22.19	19.79	18.35	15.25	23.11
No-Arbitrage $A_1(4)$ GARCH			27.03	18.14	14.72	13.24	12.60	13.03	16.46
Panel C: Unconditional Correlations with EGARCH(1, 1) Volatility									
No-Arbitrage $A_0(3)$ GARCH	0.65	0.66	0.70	0.79	0.85	0.90	0.92	0.91	0.80
JK $A_1(4)$ Model	0.57	0.60	0.66	0.67	0.64	0.60	0.56	0.46	0.60
Quadratic Model	0.65	0.66	0.70	0.70	0.68	0.65	0.62	0.54	0.65
No-Arbitrage $A_1(4)$ GARCH	0.90	0.91	0.94	0.95	0.93	0.90	0.86	0.73	0.89
Panel D: RMSEs of Volatility Based on EGARCH(1, 1) Volatility									
No-Arbitrage $A_0(3)$ GARCH	48.36	40.05	28.92	19.04	12.77	9.15	7.23	5.57	21.39
JK $A_1(4)$ Model	30.33	25.96	23.83	17.61	12.86	10.60	9.52	8.80	17.44
Quadratic Model	48.71	41.21	27.76	18.92	14.50	12.20	10.89	8.86	22.88
No-Arbitrage $A_1(4)$ GARCH	41.17	33.18	21.63	12.93	8.56	6.90	6.55	6.88	17.23
Panel E: RMSEs of Yields									
No-Arbitrage $A_0(3)$ GARCH	8.13	11.89	16.21	8.10	6.38	6.61	6.95	14.81	9.88
JK $A_1(4)$ Model	12.42	11.35	20.15	8.95	4.13	6.72	10.06	5.78	9.95
Quadratic Model	9.69	12.69	14.22	7.53	7.23	7.24	6.74	8.84	9.27
No-Arbitrage $A_1(4)$ GARCH	8.06	12.45	15.02	6.57	5.18	7.33	9.45	6.67	8.84

Notes to Table: We present the unconditional correlations, the volatility fit, and the yield fit for the no-arbitrage $A_0(3)$ model with GARCH, the Joslin and Konchitchki (2018) $A_1(4)$ model, the quadratic model, and the no-arbitrage $A_1(4)$ model with GARCH. Panel A presents the unconditional correlations between the realized volatility and the one-month conditional volatility implied by the four models. We construct monthly realized variance using within-month squared changes in daily yields. Panel C presents the unconditional correlations between the EGARCH(1, 1) volatility and the one-month conditional volatility implied by the four models. The EGARCH(1, 1) is estimated assuming that the conditional mean of changes in monthly yields is generated by an AR(1) process. Panels B and D present the volatility fit for the four models. In Panel B, the RMSEs are computed by using realized volatility as a measure of model-free volatility. In Panel D, the RMSEs are computed by using an EGARCH(1, 1) model as a measure of model-free volatility. Panel E presents RMSEs based on the yield fit for the four models. RMSEs are reported in basis points. The last column shows the averages across all maturities.

(a) Additional benchmark models



es of at-the-money implied volatility.
plot the time series of constant-maturity (90 days) at-the-money implied volatility computed using the $A_1(3)$ and no-arbitrage GARCH m
plied volatility from the data. Panel A reports on options written on five-year Treasury futures contracts. Panel B reports on options o

(b) At-the-money Treasury futures option implied volatility

5.10 Table 11 + Table 12: Option-pricing errors

Read:

- Table 11 reports pricing errors for options on five-year Treasury futures.
- The no-arbitrage GARCH model has average RMSE of 1.83 percent, compared with 2.86 percent for the $A_1(3)$ model.
- Table 12 reports pricing errors for options on ten-year Treasury futures.
- The GARCH model has average RMSE of 2.48 percent, compared with 4.05 percent for the $A_1(3)$ model.
- The improvement is visible across most maturity-moneyness buckets.

Economic message: The out-of-sample option evidence confirms that better volatility modeling has economic value beyond matching yield moments.

Table 11
Option pricing errors for options on five-year treasury futures.

Panel A: Root Mean Squared Error: No-Arbitrage GARCH

	Moneyness (K/F)						
Maturity	90–95	95–97.5	97.5–100	100–102.5	102.5–105	105–110	All
<90	3.17%	1.21%	2.24%	2.33%	1.39%	2.81%	2.18%
90–182	0.77%	1.05%	1.59%	1.71%	1.33%	1.17%	1.49%
182–270	0.42%	0.89%	1.20%	1.27%	1.31%	1.41%	1.20%
All	2.48%	1.13%	1.87%	1.98%	1.36%	2.24%	1.83%

Panel B: Root Mean Squared Error: $A_1(3)$

	Moneyness (K/F)						
Maturity	90–95	95–97.5	97.5–100	100–102.5	102.5–105	105–110	All
<90	3.94%	2.31%	3.33%	3.42%	2.47%	1.90%	3.14%
90–182	1.97%	2.25%	2.70%	2.80%	2.40%	1.73%	2.58%
182–270	1.42%	2.44%	2.76%	2.89%	2.74%	2.88%	2.76%
All	3.28%	2.29%	2.98%	3.08%	2.44%	1.84%	2.86%

Panel C: Mean Absolute Error: No-Arbitrage GARCH

	Moneyness (K/F)						
Maturity	90–95	95–97.5	97.5–100	100–102.5	102.5–105	105–110	All
<90	1.96%	1.01%	2.11%	2.19%	1.20%	1.77%	1.82%
90–182	0.55%	0.91%	1.42%	1.54%	1.18%	0.91%	1.29%
182–270	0.27%	0.73%	1.10%	1.18%	1.23%	1.36%	1.09%
All	1.38%	0.96%	1.69%	1.80%	1.19%	1.40%	1.52%

Panel D: Mean Absolute Error: $A_1(3)$

	Moneyness (K/F)						
Maturity	90–95	95–97.5	97.5–100	100–102.5	102.5–105	105–110	All
<90	2.67%	1.96%	3.03%	3.14%	2.13%	1.43%	2.68%
90–182	1.39%	1.98%	2.42%	2.54%	2.15%	1.43%	2.28%
182–270	1.30%	2.36%	2.70%	2.83%	2.67%	2.86%	2.69%
All	2.14%	1.97%	2.69%	2.81%	2.16%	1.45%	2.47%

Notes to Table: We present the root mean squared error (RMSE) and the mean absolute error (MAE) for a sample of call and put Treasury futures options. The underlying security is the five-year maturity Treasury futures contract. The sample period is from May 1988 to June 2016. Panels A and B present the RMSEs for the no-arbitrage GARCH model and the $A_1(3)$ model respectively. Panels C and D present the MAEs for these two models. The errors are computed for different moneyness and maturity groups. Moneyness ranges between 90% and 110%. The maturities are divided into three groups: less than 90 days, 90 to 182 days, and 182 to 270 days.

(a) Option-pricing errors for five-year Treasury futures

Table 12
Option pricing errors for options on ten-year treasury futures.

Panel A: Root Mean Squared Error: No-Arbitrage GARCH

	Moneyness (K/F)						
Maturity	90–95	95–97.5	97.5–100	100–102.5	102.5–105	105–110	All
<90	2.36%	3.12%	3.66%	3.69%	3.03%	1.82%	3.06%
90–182	1.59%	2.05%	2.33%	2.44%	2.36%	1.96%	2.15%
182–270	1.40%	1.78%	1.89%	2.01%	2.06%	2.17%	1.89%
All	1.91%	2.46%	2.77%	2.87%	2.59%	1.93%	2.48%

Panel B: Root Mean Squared Error: $A_1(3)$

	Moneyness (K/F)						
Maturity	90–95	95–97.5	97.5–100	100–102.5	102.5–105	105–110	All
<90	4.17%	4.49%	5.05%	5.09%	4.42%	2.73%	4.46%
90–182	3.30%	3.77%	4.00%	4.07%	3.85%	3.09%	3.74%
182–270	3.43%	3.96%	4.19%	4.27%	4.15%	3.77%	4.02%
All	3.67%	4.07%	4.39%	4.47%	4.10%	3.04%	4.05%

Panel C: Mean Absolute Error: No-Arbitrage GARCH

	Moneyness (K/F)						
Maturity	90–95	95–97.5	97.5–100	100–102.5	102.5–105	105–110	All
< 90	1.64%	2.83%	3.46%	3.50%	2.80%	1.50%	2.66%
90–182	1.30%	1.82%	2.12%	2.24%	2.16%	1.72%	1.91%
182–270	1.18%	1.54%	1.71%	1.85%	1.91%	2.03%	1.69%
All	1.42%	2.14%	2.48%	2.61%	2.36%	1.67%	2.14%

Panel D: Mean Absolute Error: $A_1(3)$

	Moneyness (K/F)						
Maturity	90–95	95–97.5	97.5–100	100–102.5	102.5–105	105–110	All
<90	3.18%	4.13%	4.65%	4.63%	3.90%	2.27%	3.85%
90–182	2.76%	3.34%	3.56%	3.62%	3.38%	2.66%	3.26%
182–270	3.17%	3.74%	4.00%	4.07%	3.96%	3.62%	3.81%
All	2.97%	3.68%	3.98%	4.03%	3.64%	2.61%	3.55%

Notes to Table: We present the root mean squared error (RMSE) and the mean absolute error (MAE) for a sample of call and put Treasury futures options. The underlying security is the ten-year maturity Treasury futures contract. The sample period is from May 1988 to June 2016. Panels A and B present the RMSEs for the no-arbitrage GARCH model and the $A_1(3)$ model respectively. Panels C and D present the MAEs for these two models. The errors are computed for different moneyness and maturity groups. Moneyness ranges between 90% and 110%. The maturities are divided into three groups: less than 90 days, 90 to 182 days, and 182 to 270 days.

(b) Option-pricing errors for ten-year Treasury futures

6 Short summary

- The paper proposes no-arbitrage term structure models with GARCH volatility factors.
- Standard affine stochastic volatility models make volatility a linear function of factor levels; the proposed model makes volatility a function of lagged squared innovations.
- The baseline empirical model has three latent yield factors and one GARCH volatility factor.
- The model is estimated on monthly Treasury yields from November 1971 to October 2019.
- Model-free volatility benchmarks are realized volatility and EGARCH(1,1) volatility.
- The GARCH model has stronger correlations with model-free volatility and lower volatility RMSEs than canonical and restricted $A_1(3)$ models.
- The improvement is largest at intermediate and long maturities and during periods with large volatility changes.
- The model's yield fit remains close to that of standard stochastic volatility term-structure models.
- The model also matches Campbell–Shiller bond-risk-premium regression patterns.
- Robustness checks with expected realized variance, multiple volatility factors, principal components, and alternative benchmark models preserve the main conclusion.
- In an out-of-sample Treasury futures option application, the GARCH model generates much smaller pricing errors than the canonical $A_1(3)$ model.
- The main lesson is that the parametric form of volatility dynamics is critical in term-structure modeling.

7 Conclusion

7.1 Core conclusion

Doshi, Jacobs, and Liu show that the poor volatility performance of standard affine term structure models is not inevitable. A tractable no-arbitrage model with GARCH volatility can fit conditional yield volatility much better while maintaining a reasonable fit to the yield curve.

7.2 Economic intuition

Yield volatility is not only related to the level, slope, and curvature of the yield curve. It also reacts to recent squared shocks to yield factors. The GARCH specification directly captures this shock-driven volatility clustering. This allows the model to fit both the high-volatility early-1980s period and later low-volatility periods more accurately.

7.3 Takeaway

The paper's practical message is that term-structure models should pay as much attention to the volatility process as to the factor structure of yields. A small change in the variance dynamic—using lagged squared innovations rather than factor levels—substantially improves volatility fit and derivative pricing.